A Appendix: matrix algebra

A.1 Matrix Transpose

The transpose of a matrix A, denoted A^T, is obtained by flipping the matrix over its diagonal (switching rows and columns). If A is an $m \times n$ matrix:

$$(A^T)_{ij} = A_{ji}$$
$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, \quad A^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix} \tag{A.1}$$

Properties:

- $(A + B)^T = A^T + B^T$
- $(AB)^T = B^T A^T$ (Order reverses)
- $(A^{-1})^T = (A^T)^{-1}$

A.2 Determinant

The determinant $\det(A)$ or $|A|$ is a scalar value defined only for square matrices ($n \times n$). It represents the volume scaling factor of the linear transformation.

- **For a 2×2 matrix:**

$$\det\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = ad - bc \tag{A.2}$$

- **For a 3×3 matrix (Sarrus Rule):**

$$\det\left(\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}\right) = a(ei - fh) - b(di - fg) + c(dh - eg) \tag{A.3}$$

Robotics Significance: if $\det(J) = 0$, where J is the Jacobian, the robot is in a **singularity**.

A.3 Matrix Inverse

For a square matrix A, the inverse A^{-1} satisfies $AA^{-1} = A^{-1}A = I$, where I is the identity matrix. The inverse exists **if and only if** $\det(A) \neq 0$ (the matrix is non-singular).

$$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A) \tag{A.4}$$

Where $\operatorname{adj}(A)$ is the adjugate matrix (transpose of the cofactor matrix). **Special case: rotation matrices ($R \in SO(3)$)**: rotation matrices are orthonormal, meaning:

$$R^{-1} = R^T \tag{A.5}$$

This property is used constantly to simplify kinematic equations.

A.4 Rank and singularity

The **rank** of a matrix is the maximum number of linearly independent rows or columns.

- **full rank**: an $m \times n$ matrix has full rank if its rank equals $\min(m, n)$. For a square matrix, full rank means it is invertible (non-singular);
- **losing rank**: a matrix "loses rank" when its rows or columns become linearly dependent.

Robotics Significance: when a Jacobian $J(q)$ loses rank:

1. the robot loses the ability to move in at least one direction in Cartesian space (mobility loss);
2. small desired velocities in Cartesian space may require infinite velocities in joint space ($\dot{q} = J^{-1}\dot{x}$ involves division by a determinant approaching zero).

A.5 Skew-Symmetric matrices

A matrix $S$is skew-symmetric if $S^T = -S$. In robotics, we use the operator $[\omega]_\times$ to represent a cross product as a matrix multiplication: to represent a cross product as a matrix multiplication:

$$\omega \times v = S(\omega)v = \begin{bmatrix} 0 & -\omega_z & \omega_y \, \omega_z & 0 & -\omega_x & -\omega_y & \omega_x & 0 \end{bmatrix} \begin{bmatrix} v_x \; v_y \; v_z \end{bmatrix} \tag{A.6}$$

This is fundamental for defining link velocities and the derivative of rotation matrices: $\dot{R} = S(\omega)R$.

A.6 Positive definiteness

A symmetric matrix M is **positive definite** ($M > 0$) if

$$x^T M x > 0 \tag{A.7}$$

for all non-zero vectors x.

Robotics Significance: the inertia matrix $M(q)$ must always be positive definite. This ensures that the kinetic energy

$$T = \frac{1}{2}\dot{q}^T M(q)\dot{q}$$

is always positive for any non-zero motion. A practical check is that all eigenvalues of M must be strictly positive.

A.7 Useful identities

- **Trace**: $\mathrm{tr}(A) = \sum A_{ii}$ (sum of diagonal elements);
- **matrix differentiation**: $\frac{d}{dt}(A^{-1}) = -A^{-1}\dot{A}A^{-1}$
- **Jacobian property**: $\dot{x} = J(q)\ddot{q} \implies \ddot{x} = J(q)\ddot{q} + \dot{J}(q)\dot{q}$

A.8 Orthonormal matrices

Orthonormal matrices have the following properties:

- inverse is transpose,
- the columns of an orthonormal matrix are unit vectors that are orthogonal to each other, and the same holds true for the rows,
- determinant is always ± 1,
- multiplying a matrix Q by its transpose results in the identity matrix.

If a matrix is symmetric and square, it has eigenvalues which are real, but not necessarily all positives; if a square matrix is symmetric and positive-definite, all its eigenvalues are real and positive.

Regarding the matrix Σ, we can easily see its rank by counting how many σ_i are not 0 (given that they are oredered from larger to smallest).

A.9 Eigenvalues and eigenvectors

Un robotics, we focus on square matrices $A \in \mathbb{R}^{n \times n}$. The goal is to find scalars λ (eigenvalues) and non-zero vectors v (eigenvectors) such that:

$$Av = \lambda v \tag{A.8}$$

A.9.1 Finding eigenvalues

To find the eigenvalues, we rearrange the equation into a homogeneous system:

$$(A - \lambda I)v = 0 \tag{A.9}$$

For a non-zero vector v to exist, the matrix $A - \lambda I$ must be singular, meaning its determinant must be zero. This gives us the **characteristic equation**:

$$\det(A - \lambda I) = 0 \tag{A.10}$$

This determinant is a polynomial in λ of degree n. Solving for the roots of this polynomial gives you the eigenvalues $\lambda_1, \lambda_2, ..., \lambda_n$.

A.9.2 Finding eigenvectors

For each eigenvalue λ_i found in the previous step, you must find the corresponding vector v_i by solving the linear system:

$$(A - \lambda_i I)v_i = 0 \tag{A.11}$$

Since $A - \lambda_i I$ is singular by construction, there will be at least one free variable, meaning eigenvectors are defined up to a scaling factor. In robotics, we usually normalize them so that $\|v_i\| = 1$.

A.9.3 Example: $2(\times)2$ matrix

Given

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \tag{A.12}$$

1. solve characteristic equation:

$$(a - \lambda)(d - \lambda) - bc = 0 \implies \lambda^2 - (a + d)\lambda + (ad - bc) = 0 \tag{A.13}$$

2. use the quadratic formula to find

$$\lambda_{1,2} = \frac{(a + d) \pm \sqrt{(a + d)^2 - 4(ad - bc)}}{2} \tag{A.14}$$

3. substitute λ_1 and λ_2 back into

$$\begin{bmatrix} a - \lambda & b \\ c & d - \lambda \end{bmatrix} \begin{bmatrix} v_x \\ v_y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \tag{A.15}$$

   to find the ratio between v_x and v_y.

A.9.4 Why is this used in Robotics?

- **Inertia matrix analysis**: since the inertia matrix $M(q)$ is symmetric and positive definite, all its eigenvalues are real and positive. The eigenvectors of $M(q)$ represent the **principal axes of inertia** for the robot arm in that configuration;
- **manipulability**: for the Jacobian J, we look at the eigenvalues of JJ^T. The square roots of these eigenvalues are the **Singular Values** (σ_i). The direction of the eigenvector corresponding to the smallest eigenvalue tells you the direction in which the robot is closest to a **singularity** (the direction where it is hardest to move);
- **control stability**: in linear control theory ($\dot{x} = Ax + Bu$), a system is **asymptotically stable** if and only if all eigenvalues of the closed-loop system matrix have negative real parts.

A.9.5 Useful property: trace and determinant

For any square matrix A:

- $\det(A) = \lambda_1 \cdot \lambda_2 \cdots \lambda_n$;
- $\mathrm{tr}(A) = \lambda_1 + \lambda_2 + \cdots + \lambda_n$.